

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

Semester – I & II University Examination May 2014 (ME/CL/EE)

IT 101 Fundamentals of Computer Programming

Date: 27/05/2014, Tuesday

Time: 01:30 P.M to 04:30P.M

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1 Answer the following questions.

[11]

1. Draw the structure of C Programming and explain in brief. 04
2. Convert 10110 binary numbers into its equivalent decimal. 02
3. Fill in the Blanks with appropriate word:
 - a) Every C program must be end with a _____. 02
 - b) The _____ specification is used to read or write long integer. 03
4. State True or False:
 - a) Syntax error will be detected by the compiler. 03
 - b) Global variables are visible in all blocks and functions in the program.
 - c) Every array starts with index 0.

Q.2 Answer the following questions.

[12]

1. What are modifiers? List out all the modifiers and explain in detail.
2. What is typedef? Explain with suitable example.
3. What is enumerated data type? Explain with suitable example.

OR

Q.2 Answer the following questions.

[12]

1. Classify valid and invalid variable name in C. 04
\$ANSI, Max_Val, average, smart_phone, No.1, 13saath
2. Differentiate: printf() vs. scanf(). 04
3. Give the output of the following codes. 04
 1. void main()
{
int x, y=10;
char z='c'-'a';
x = y + z;
printf("%d", x);
}
 2. void main()
{
int i=-2;
printf("%d", -i);
}

Q.3 Answer the following questions.

[12]

1. Write a program to print the following triangle. 06

1 2
1 2 3

2. Write a program to evaluate the series: $2 + 4 + 6 + 8 + \dots + n$ 06

OR

Q.3 Answer the following questions.

[12]

1. Write a program to print the following triangle. 06

1
2 2
3 3 3

2. Write a program to evaluate the series: $1 + 3 + 5 + 7 + \dots + n$ 06

SECTION – II

Q.4 Answer the following questions. [11]

1. Explain switch case with suitable example. 03
2. What is a pointer? How can it be initialized? What is the scale factor of a pointer? 03
3. State True or False: 03
 - a) The keyword *void* is a data type in C.
 - b) The *scanf* function can be used to read only one variable at a time.
 - c) A function in C should have at least one argument.
4. Fill in the Blanks with Appropriate word: 02
 - a) _____ is the largest value that can be stored in an unsigned char variable.
 - b) A *for* loop with no test condition is also known as _____ loop.

Q.5 Answer the following questions. [12]

1. Differentiate: Exit controlled loop vs. Entry controlled loop
2. Write a program to add 3×3 matrices using Array.
3. Write a program to print the numbers of Fibonacci series up to N numbers.

OR

Q.5 Answer the following questions. [12]

1. What is data type? List out all types of data type in C. Explain each in brief.
2. Write a program to find out all the numbers divisible by 7 between 1 and 100.
3. Write a program to reverse a given string without *string.h*.

Q.6 Answer the following questions. [12]

(a) Give the output of the following codes. 04

1. void main() { int a=10; int *p; p=&a; printf("%d",*p); }	2. void main() { int count=5; while(count>0) printf("count"); }
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(b) Attempt any Two. 08

1. Explain the relational and conditional operator with example.
2. Write a syntax of *elseif* ladder with example.
3. Explain *strrev()* and *strcmp()* functions with example.
